

Part: **3.1 – STRENGTH OF METALLIC PARTS**
Chapter: **3146**
Title: **TUBES**
Revision: **0.7**
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1 INTRODUCTION

1.1 OBJECT

This chapter is concerned with the strength of circular tubes with a combination of axial, shear, bending and torsion loads simultaneously applied.

1.2 METHOD APPLICABILITY

Analysis methodologies described in the present chapter are only applicable for the structural elements which have the following characteristics:

- ✓ Straight thin-walled tubes (typically with $D/t \geq 20$) made of metallic materials with uniform circular cross-section (constant cross-section shape along beam-column's length).
- ✓ Structural tubes with loads locally applied in combined axial tension, axial compression, transversal shear, bending moments and/or torsion moment.

1.3 TUBES FAILURE MODES

A tube is a straight structural element which can be defined as a straight beam of hollow circle section (usually uniform), with a combination of axial, shear, bending and torsion loads simultaneously applied. Bending can be caused by the application of eccentric axial loads (not necessarily *centred*, not applied in tube's cross-section centroid), transversal loads or due to an initial curvature, or due to a combination of any ones of the mentioned causes.

The two principal failure modes of tubes are the following ones:

- ✓ **Buckling failure**: If compression load caused by any load state is higher than its critical buckling load. Buckling failure may appear in tube's section (local buckling) or affect to the whole tube (column buckling).

- ✓ Strength failure: If level of internal stresses caused by the combined action of the different types of loads is higher than the tube's material allowable values.

Then, applied axial loads may cause the global buckling failure of the structural element and tube's strength failure may occur if deformation at element's extreme fibre exceeds the material ultimate strength; so, both mentioned failure modes must be analysed. If the tube is only axially loaded in tension, *strength* is the only failure mode that must be analysed.

2 METHOD DESCRIPTION

The following paragraphs describe methodologies for calculation of both failure modes of tubes, with applied axial loads, shear loads, bending moment and torsion moment loads.

The following main hypotheses are considered for those analyses:

- ✓ Tube does not have any intermediate support; supports are considered to be only located at the tube ends.
- ✓ Tube is generally considered to work in *plastic regime*, however analysis method considered for beam-column effects do not cover material plasticity properties, so obtained results could not be conservative in extreme design configurations or too high loaded structures.

2.1 ALLOWABLE STRENGTHS

2.1.1 Axial tensile strength allowable

Ultimate pure tension strength of a metallic tube under applied tensile axial loads can be calculated according to the following expression:

$$P_{t_allow} = F_{tu} \cdot \frac{\pi}{4} \cdot (D^2 - d^2)$$

where:

- P_{t_allow} is the pure axial tension allowable load [N]
- F_{tu} is the tube material Ultimate Tensile Stress [MPa]
- D is the tube outer diameter [mm]
- d is the tube inner diameter [mm]

2.1.2 Axial compression local buckling & strength allowable

Ultimate pure compression strength of a metallic tube under applied compressive axial loads can be calculated according to the following expression:

$$P_{c_allow_1} = F_{cy} \cdot \frac{\pi}{4} \cdot (D^2 - d^2)$$

where:

- $P_{c_allow_1}$ is the pure axial compression allowable load [N]
- F_{cy} is the tube material Yield Compression Stress [MPa]
- D is the tube outer diameter [mm]
- d is the tube inner diameter [mm]

According to Table 15.2-15 of Ref. 2, crippling local buckling allowable of a thin-walled metallic tube under applied compressive axial loads can be calculated according to the following expression:

$$P_{c_allow_2} = 0.3 \cdot \frac{E \cdot t}{R_m} \cdot \frac{\pi}{4} \cdot (D^2 - d^2)$$

where:

- $P_{c_allow_2}$ is the crippling allowable load [N]
- E is the tube material Young Modulus [MPa]
- t is the tube wall thickness [mm]
- R_m is the tube (mid-thickness) radius [mm]
- D is the tube outer diameter [mm]
- d is the tube inner diameter [mm]

Then, tube's final local buckling & strength allowable value under axial compressive loads would be the following one:

$$P_{c_allow} = \min(P_{c_allow_1}, P_{c_allow_2})$$

Notice that global buckling failure mode of the tube under axial compression loads is not covered by previous expressions since it is independent to the local failure modes above mentioned, so column buckling (global buckling) must be separately analysed and not combined with other failure modes. See section 2.3 for more information.

2.1.3 Bending buckling & strength allowable

The maximum allowable bending moment of a circular tube with constant thickness M_{allow} is determined from its geometrical data and material properties and can be obtained using the equations listed in chapter 3 of Ref. 5. Those equations cover the following failure modes of the tube: rupture and local buckling.

$$M_{b_allow} = F_b \cdot W_b = F_b \cdot \frac{\pi}{32} \cdot \frac{D^4 - (D - 2t)^4}{D}$$

$$F_b = F_{b_o} \cdot \left[1 - 0.3027 \cdot \left(\frac{F_{b_o}}{F_{cr_e}} \right)^{0.75} \right] \quad ; \quad F_{cr_e} = 0.8144 \cdot E \cdot \frac{\left(\frac{D}{t} \right)^2}{\left(\frac{D}{t} \right)^3 - 3 \cdot \left(\frac{D}{t} \right)^2 + 4 \cdot \frac{D}{t} - 2}$$

$$\text{if } F_b \geq 2.3333 \cdot F_{cr_e} \quad \text{then} \quad F_b = F_{cr_e}$$

where:

- M_{b_allow} is the allowable bending moment at failure [N·mm]
- W_b is the tube cross-section bending modulus [mm³]
- E is the tube material Young Modulus [MPa]
- D is the tube outer diameter [mm]
- t is the tube wall thickness [mm]
- F_b is the Bending Modulus of Rupture [MPa]
- F_{b_o} is the basic value of F_b [MPa]
- F_{cr_e} is the elastic critical buckling stress [MPa]

Basic value F_{b_o} of Bending Modulus of Rupture can be obtained with the following expressions, according to *Cozzone* method (see section C3.4 of Ref. 1):

$$F_{b_o} = F_{tu} + (K - 1) \cdot F_o \quad ; \quad K = \frac{4D}{3\pi t} \cdot \frac{3 \cdot \left(\frac{D}{t}\right)^2 - 6 \cdot \frac{D}{t} + 4}{\left(\frac{D}{t}\right)^3 - 3 \cdot \left(\frac{D}{t}\right)^2 + 4 \cdot \frac{D}{t} - 2}$$

where:

- F_{tu} is the tube material Ultimate Tensile Stress [MPa]
- K is the tube cross-section geometrical shape factor.
- F_o is the *Cozzone*'s fictitious stress at the neutral axis or at zero strain location [MPa]

Bending Modulus of Rupture F_b is also provided for specific materials and tubes arrangements in Figures C4.11 to C4.14 of Ref. 1. This F_b value already covers local buckling effects.

2.1.4 Transversal shear strength allowable

According to Table A14.2 of Ref. 1, ultimate pure shear strength of a metallic tube under applied transversal shear loads can be calculated according to the following expression:

$$Q_{s_allow} = \frac{0.75 \cdot F_{su} \cdot A}{1 + \frac{D \cdot d}{D^2 + d^2}}$$

where:

- Q_{s_allow} is the pure shear allowable load [N]
- F_{su} is the tube material Ultimate Shear Stress [MPa]
- A is the tube cross-section effective area [mm²] $\rightarrow A = \frac{\pi}{4} \cdot (D^2 - d^2)$

- D is the tube outer diameter [mm]
 d is the tube inner diameter [mm]

The methodology described in this SMM chapter is only applicable to slender tubes. The risk of buckling due to pure shear loads is not significant compared with other buckling modes (caused by axial compression loads, bending moments or torsion moments).

2.1.5 Torsion buckling & strength allowable

Ultimate pure torsion strength of a metallic tube under applied torsional moment can be calculated according to the following expression:

$$M_{tor_allow_1} = \frac{F_{su} \cdot J}{R}$$

where:

- $M_{tor_allow_1}$ is the pure torsion allowable torsional moment [N·mm]
 F_{su} is the tube material Ultimate Shear Stress [MPa]
 J is the area polar moment of inertia of tube's cross-section [mm⁴] $\rightarrow J = \frac{\pi}{2} (R^4 - r^4)$
 R is the tube cross-section outer radius [m]
 r is the tube cross-section inner radius [m]

Buckling elastic shear allowable stress due to pure torsion loads in elastic regime, of a thin-walled circular tube with constant thickness, can be obtained according to the following expression, extracted from chapter 2 of Ref. 3:

$$\tau_{tor_allow} = E \cdot K \cdot \left(\frac{t}{R}\right)^\alpha \left(\frac{R}{L}\right)^\beta \rightarrow M_{tor_allow_2} = \frac{\tau_{tor_allow} \cdot J}{R}$$

where:

- $M_{tor_allow_2}$ is the buckling elastic allowable torsional moment [N·mm]
 τ_{tor_allow} is the torsion elastic shear allowable stress [N/m²]
 E is the tube material Young Modulus [N/m²]
 L is the tube length [m]
 t is the tube wall thickness [m]
 K, α, β are the following experimentally obtained values: $K = 0.292$ $\alpha = 3/2$ $\beta = 0$
Ref. 3 also provides higher values of α, β and K for tubes with low L/R , if needed.

According to section C4.19 of Ref. 1, allowable torsion moments are provided for several specific materials and tubes arrangements in Figures C4.17 to C4.31 of Ref. 1.

Then, tube's final buckling & strength allowable value under torsion moment would be the following one:

$$M_{tor_allow} = \min(M_{tor_allow_1}, M_{tor_allow_2})$$

2.2 FAILURE MODES INTERACTION CRITERION

Reserve factors for combined loads in tubes with $D/t \leq 100$, axially loaded in compression may be calculated according to the equation shown in sub-chapter 3.2.5 of Ref. 4:

$$RF_{combined} = \frac{1}{R_c + R_{tor} + \sqrt{R_b^2 + R_s^2}}$$

In the other hand, for tubes with $D/t \leq 100$, axially loaded in tension, the combined reserve factor may be calculated according to the equations shown in sub-chapters 3.2.1, 3.2.3 and 3.2.4 of Ref. 4:

- Tension and Bending: $R_t^{1.5} + R_b = 1 \quad (*)$

- Tension and Shear: $R_t^2 + R_s^3 = 1 \quad (*)$

- Tension and Torsion: $R_t^2 + R_{tor}^2 = 1 \quad \rightarrow \quad RF_{combined} = \frac{1}{\sqrt{R_t^2 + R_{tor}^2}}$

- All combined loads: $R_t^{1.5} + R_b + R_s^3 + R_{tor}^2 = 1 \quad (*)$

(*) $RF_{combined}$ should be obtained by iterative numerical methods. Replace in equations above the failure indices by $R_i = RF \cdot P_i / P_{i_allow}$, where P_i is the applied load or moment and P_{i_allow} the corresponding (uniaxial) allowable load or moment, and then numerically solve to obtain the RF .

where:

R_t is the ratio of the applied tensile load to its allowable value ($R_t = P_t / P_{t_allow}$)

R_c is the ratio of the applied compression load to its allowable value ($R_c = P_c / P_{c_allow}$)

R_b is the ratio of the applied bending moment to its allowable value ($R_b = M_b / M_{b_allow}$) (**)

R_s is the ratio of the applied transversal shear load to its allowable value ($R_s = Q_s / Q_{s_allow}$)

R_{tor} is the ratio of the applied torsion moment to its allowable value ($R_{tor} = M_{tor} / M_{tor_allow}$)

(**) The bending moment M_b applied to the tube must be increased in case of tubes axially loaded in compression due to its beam-column behaviour. See section 2.4 for further details about beam-column effects and how tube's maximum bending moment must be calculated in these cases.

Note that as stated in sub-chapter 3.3 of Ref. 4, if one or more stresses are zero, then the corresponding R ratios are also zero, and the equation given above is still valid.

Also note that all mentioned allowable loads & stresses from above equations should be the minimum values between pure strength and buckling allowable values where applicable for each separated failure mode.

2.3 COLUMN BUCKLING ANALYSIS

Column buckling phenomenon is a global failure of the tube, independent to other local failure modes caused by axial compression loads, so it must be separately analysed and not combined with other failure modes.

Column buckling analysis methods described in Airbus DS SMM Chapter 3140 'Column Buckling in Compression' are also applicable for metallic tubes to obtain their corresponding column-buckling allowable value (P_{col_allow}).

Then, column buckling Reserve Factor must be calculated with the following expression, where P_c is the axial compression load applied to the tube:

$$RF_{col} = \frac{P_{col_allow}}{P_c}$$

2.4 TUBE BEAM-COLUMN ANALYSIS

Beam-column analysis methods described in Airbus DS SMM Chapter 3145 'Beam-Columns' are also applicable for metallic tubes.

Equations provided for subcases #5 & #6 in Table 1 of Airbus DS SMM Chapter 3145 may be used to obtain the maximum bending moment at a metallic tube under applied bending moments at its ends and axial compression loads. However, it should be noted that considered beam-column analysis method does not cover material plasticity effects, so obtained results could not be conservative in extreme design configurations or high loaded structures.

2.5 PLASTICITY EFFECT IN BEAM-COLUMN

Reserved for future work.

2.6 CURVED TUBES ANALYSIS

Reserved for future work.

2.7 WELDED TUBES ANALYSIS

Reserved for future work.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
D	mm	Tube outer diameter
RF		Reserve Factor
SMM		Stress Methods Manual
t	mm	Tube wall thickness

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E.F.Bruhn]	1973	“Analysis and Design of Flight Vehicle Structures”.
Ref. 2	[W.C.Young, R.G.Budynas]	7 th Ed.	“Roark’s Formulas for Stress and Strain”
Ref. 3	ESDU 84012	-	Buckling of thin-walled unstiffened circular cylinders under combined torsion and pressure.
Ref. 4	H.S.B. 51200-01	G	Reserve factors from interaction curves for combined loading.
Ref. 5	H.S.B. 52130-01	D	Allowable bending moment of circular tubes.

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
SAM (Airbus DS)	1.3 (05/2023)	Airbus DS experimental stress software (SAM standing for Structural Analysis Methods) including an implementation of the tubes sizing methodology described in this report

Table 1. Software programs referenced in this chapter.

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.7 17/May/2023	G. Moreda (TEPAP-TL4)	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected.

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEPAP-TL4)</i>	Gustavo Baños DD/MM/AAAA	Florian Glock DD/MM/AAAA

Table 3. *Approval signatures table for last revision of this chapter.*

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